

MAT 213 Brief Calculus

Name: _____

Mock Midterm C

Date: _____

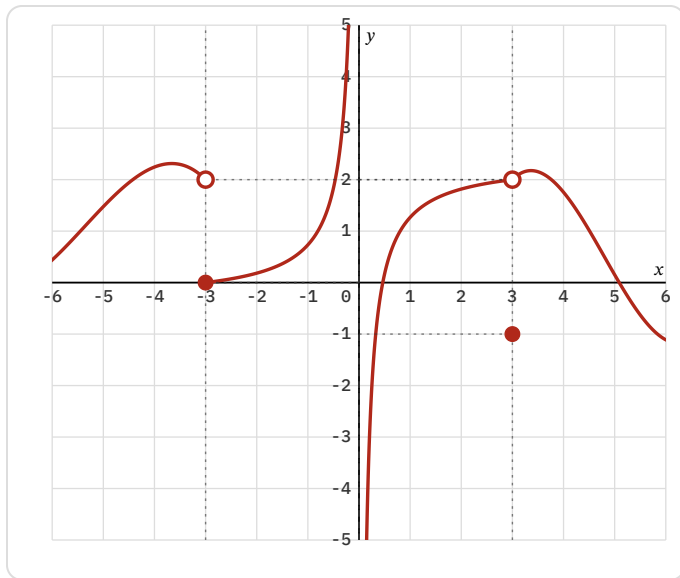
Practice midterm (AI-Free) · Sections 1.1–1.4

Score: _____ / 88

Directions: Answer each question below. Be sure to show work to receive full credit. Work this exam on your own, without notes or AI, the way you will on the proctored midterm. Be sure the notation in your solutions matches the notation from the lesson videos. Write **DNE** if a value or limit does not exist. Suggested time: 75 minutes.

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write **DNE**.



a.) $f(0)$ _____ (1)

b.) $\lim_{x \rightarrow 0^-} f(x)$ _____ (1)

c.) $\lim_{x \rightarrow 0^+} f(x)$ _____ (1)

d.) $\lim_{x \rightarrow 0} f(x)$ _____ (1)

e.) $f(3)$ _____ (1)

f.) $\lim_{x \rightarrow 3^-} f(x)$ _____ (1)

g.) $\lim_{x \rightarrow 3^+} f(x)$ _____ (1)

h.) $\lim_{x \rightarrow 3} f(x)$ _____ (1)

8 pts

2. Let $f(x) = \frac{1}{(x+2)^2}$. Answer with a number, ∞ , $-\infty$, or DNE.

5 pts

a.) $\lim_{x \rightarrow -2^-} f(x)$ _____ (1)

b.) $\lim_{x \rightarrow -2^+} f(x)$ _____ (1)

c.) $\lim_{x \rightarrow -2} f(x)$ _____ (1)

d.) $\lim_{x \rightarrow \infty} f(x)$ _____ (1)

e.) The vertical asymptote is $x =$ _____ (1)

3. Suppose

4 pts

$$\lim_{x \rightarrow 3} f(x) = 4, \quad \lim_{x \rightarrow 3} g(x) = 2, \quad \lim_{x \rightarrow 3} h(x) = 9$$

Use the properties of limits to find each limit.

a.) $\lim_{x \rightarrow 3} [3f(x) + g(x)]^2$ _____ (1)

b.) $\lim_{x \rightarrow 3} ([f(x)]^3 - g(x))$ _____ (1)

c.) $\lim_{x \rightarrow 3} \sqrt{h(x)} f(x)$ _____ (1)

d.) $\lim_{x \rightarrow 3} [3f(x) - 4g(x)]$ _____ (1)

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{x \rightarrow -3} \frac{3x^2 + 6x - 9}{3x^2 + 14x + 15}$ _____ (2)

b.) $\lim_{x \rightarrow 9} \frac{x - 9}{\sqrt{x} - 3}$ _____ (2)

4 pts

5.

Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results.

6 pts

a.) $\lim_{x \rightarrow -\infty} \frac{5}{-3x^3 + 4x^2}$ _____ (2)

b.) $\lim_{x \rightarrow \infty} \frac{-2x^3 - 4x^2}{-x + 6}$ _____ (2)

c.) $\lim_{x \rightarrow -\infty} \frac{3x + 2}{4x + 9}$ _____ (2)

6. Find each limit (end behavior). Use ∞ or $-\infty$ when the outputs grow without bound. 4 pts

a.) $\lim_{x \rightarrow \infty} (3x^5 + 3x^4 - 3x^2 - 3)$ _____ (1)

b.) $\lim_{x \rightarrow \infty} (4 - 4e^x)$ _____ (1)

c.) $\lim_{x \rightarrow -\infty} (-1 + \frac{1}{x^2})$ _____ (1)

d.) $\lim_{x \rightarrow \infty} (-x^5 + x^2 - 6x)$ _____ (1)

7. Let 6 pts

$$f(x) = \begin{cases} \frac{x^2 + x - 6}{x + 3} & x < -3 \\ 2x + 5 & x > -3 \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 1} f(x)$ _____ (1)

b.) $\lim_{x \rightarrow -6} f(x)$ _____ (1)

c.) $f(-3)$ _____ (1)

d.) $\lim_{x \rightarrow -3^-} f(x)$ _____ (1)

e.) $\lim_{x \rightarrow -3^+} f(x)$ _____ (1)

f.) $\lim_{x \rightarrow -3} f(x)$ _____ (1)

8.

Find the value of k that makes f continuous at $x = 2$.

4 pts

$$f(x) = \begin{cases} x^2 - 2x & x < 2 \\ kx - 1 & x \geq 2 \end{cases}$$

a.) $\lim_{x \rightarrow 2^-} f(x)$ _____ (2)

b.) $k =$ _____ (2)

9.

Classify each function at the given x -value.

3 pts

a.) $f(x) = \begin{cases} x & x < 3 \\ x - 2 & x \geq 3 \end{cases}$ at $x = 3$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

b.) $f(x) = \frac{x-3}{x^2-2x-3}$ at $x = -1$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

c.) $f(x) = \frac{x+1}{x^2-x-2}$ at $x = -1$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

10.

Let $C(q) = 2q^2 + 14q + 67$, where $C(q)$ represents the cost, in dollars, of producing q items. Find the average rate of change over $[1, 3]$.

a.) $C(1)$ _____ (2)

b.) $C(3)$ _____ (2)

c.) Average rate of change (number only) _____ (2)

d.) Which sentence correctly states the result **with units**? (1)

(A) Between $q = 1$ and $q = 3$, the cost changes at an average rate of **22** items per dollar.

(B) At exactly $q = 3$ items, the cost is changing at a rate of **22** dollars per item.

(C) The cost changes by a total of **22** dollars from $q = 1$ to $q = 3$.

(D) Between $q = 1$ and $q = 3$ items, the cost changes at an average rate of **22** dollars per item.

7 pts

11. The table shows the average price of a gallon of regular gas.

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Price, \$	2.37	2.37	2.22	2.10	1.98	1.92	2.08	2.27	2.21	2.45	2.57

Rounded answers (2+ decimal places) are accepted.

- a.) Average rate of change from 2016 to 2020 _____ (2)
- b.) Average rate of change from 2023 to 2026 _____ (2)
- c.) The units of these answers are... (1)

(A) dollars per year

(B) years per dollar

(C) dollars

(D) dollars per gallon

5 pts

12.

Let $f(x) = x^3 - 2x$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.

a.) Simplified difference quotient (in terms of x and h) _____ (3)

b.) $x = 1, h = 1$ _____ (1)

c.) $x = 1, h = 0.1$ _____ (1)

d.) $x = 1, h = 0.01$ _____ (1)

e.) As $h \rightarrow 0$, what value do these approach? _____ (1)

7 pts

13.

Let $f(x) = \sqrt{x}$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ _____ (3)

b.) $f'(4) =$ _____ (2)

c.) Slope of the tangent line to f at $x = 9$ _____ (2)

d.) What does $f'(4) = \frac{1}{4}$ mean? (1)

(A) The slope of the secant line from $x = 4$ to $x = 4 + 1$ is $\frac{1}{4}$.

(B) The instantaneous rate of change of f at $x = 4$ is $\frac{1}{4}$ — the slope of the tangent line there.

(C) The average rate of change of f from $x = 0$ to $x = 4$ is $\frac{1}{4}$.

(D) The value of f at $x = 4$ is $\frac{1}{4}$.

8 pts

14. (a) Suppose $f(1) = -5$ and $f'(1) = -5$. Write the equation of the tangent line to f at $x = 1$.
- (b) The tangent line to the graph of g at $x = 2$ is $y = -3x + 3$. Find $g(2)$ and $g'(2)$.
- a.) (a) Tangent line _____ (3)
- b.) (b) $g(2)$ _____ (2)
- c.) (b) $g'(2)$ _____ (2)

7 pts

15.

Let $T(t) = -t^2 + 30t + 70$ be the temperature, in °F, of an oven t minutes after it is turned on.

a.) $T'(t) =$ _____ (3)

b.) $T'(10)$ _____ (2)

c.) Interpret $T'(10) = 10$. (1)

(A) Over the first 10 minutes, the oven's temperature changed at an average rate of 10 degrees Fahrenheit per minute.

(B) At $t = 10$ minutes, the oven's temperature is 10 degrees Fahrenheit.

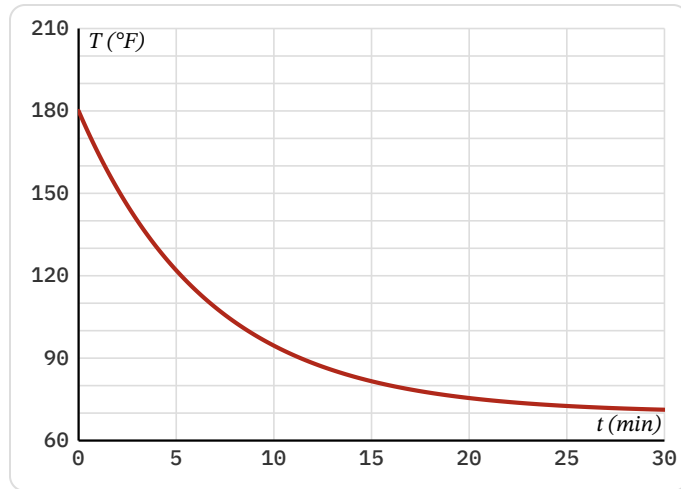
(C) At $t = 10$ minutes, the oven's temperature is increasing at a rate of 10 degrees Fahrenheit per minute.

(D) At $t = 10$ minutes, the oven's temperature is decreasing at a rate of 10 degrees Fahrenheit per minute.

6 pts

16.

A cup of coffee is left on a counter. Its temperature T (in $^{\circ}\text{F}$) after t minutes is graphed below. Decide whether each statement is true or false.



- a.) T F The instantaneous rate of change at $t = 25$ is closer to 0 than the average rate of change over $[0, 10]$. (1)
- b.) T F The coffee is cooling faster at $t = 3$ minutes than at $t = 20$ minutes. (1)
- c.) T F The tangent lines get steeper as t increases. (1)
- d.) T F The average rate of change over $[0, 10]$ is positive. (1)

4 pts
